# A Measure-Theoretic Discrete Curvature Framework for Metabolic Gene Sensitivity: From Active-Set Geometry to Transcriptional Response

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**Abstract**

Parametric flux balance analysis defines, for every metabolic network, a piecewise-affine map from environmental and genetic parameters to optimal fluxes. We show that the distributional second derivative of this map is a matrix-valued Radon measure concentrated on the codimension-one strata of the active-set complex, and that this measure — not a smooth curvature — is the canonical carrier of the network's rerouting geometry: loop holonomy scales linearly in loop size (slope 1.00, versus the quadratic scaling of any smooth map), and the second-order response mass concentrates (93.4–100.0% per boundary-crossing sweep) on operational active-set switches. We prove a refinement–resolution bridge: under mesh refinement the atomic measure converges weakly to the smooth curvature density, the dual-cell reconstruction converges strongly in $L^1$, and an exact counterexample calculus shows that total-variation convergence fails in general; a resolution parameter $\sigma$ separates the discrete and smooth regimes through the measured window $h \ll \sigma \ll L_{\mathrm{var}}$ ($L_{\mathrm{var}}$: the variation scale of the smooth object). From the measure we derive a gene-level sensitivity metric $\kappa^{\mu}$ (measure mass along a parametric trajectory) and show, on *E. coli* (M3D microarrays, $n = 424$ genes), that $\kappa^{\mu}$ predicts carbon-depletion transcriptional response ($r = +0.395$, $p = 2.6 \times 10^{-17}$; partial $r = +0.269$ controlling reference level; metric-invariant $\rho = 0.99998$ against the lexicographic definition; tie-break-robust across four alternative selection rules, and the underlying event measures stabilize at the Glivenko–Cantelli rate across random panels), while the association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation. Double-knockout epistasis aligns with active-set structure ($\rho_S = 0.865$), and genotype loops close with phenotypic memory (66% non-reverting). The framework unifies three previously distinct sensitivity objects as one measure at different resolutions.

**Keywords:** discrete curvature; active set; parametric linear programming; flux rerouting; transcriptional response; carbon depletion

## 1 Introduction

**The object.** Let a metabolic network with $m$ reactions be operated by flux balance analysis (Varma and Palsson, 1994; Orth et al., 2010) with parameters $\theta \in \Theta \subset \mathbb{R}^p$ (uptake bounds, knockdown scalings) entering the constraints affinely. The optimal flux map $v : \Theta \to \mathbb{R}^m$ is continuous and piecewise affine (parametric linear programming (Borrelli et al., 2003)), with breakpoints on the active-set strata of the constraint complex. Its distributional second derivative $\mu = D^2 v$ is therefore not a function but a *measure* — the discrete curvature carrier of the network's rerouting geometry.

**The claim structure.** This manuscript establishes, in order:

(i) the atom structure and weak convergence of $\mu_h$ under refinement (Theorem 3.1, §3), with an exact counterexample calculus delimiting what is provable (weak convergence, strong cell-wise reconstruction) from what fails (total-variation convergence);

(ii) the two-layer separation: the codimension-one Hessian layer is the unbiased curvature estimator; the codimension-two angle-defect layer converges to the Gauss-map area density with an exact sec-law bias (Proposition 3.3);

(iii) the regime dichotomy: piecewise-affine maps with fixed kink complexes produce $O(\varepsilon)$ loop holonomy (measured slope 1.00) where smooth maps produce $O(\varepsilon^2)$;

(iv) the value function $\Phi = c_{\text{bio}}^{\top} v^*$ is the canonical tie-break-free carrier, with shadow-price jumps measurable to 6–7 digits against Danskin envelopes;

(v) the measure-derived metric $\kappa^{\mu}$ predicts carbon-depletion transcriptional response ($r = +0.395$), generalizes to oxygen-limitation and carbon-switch trajectories with matched responses ($r = +0.32, +0.22$; §5.5), is robust to the tie-break convention: $r \in [+0.386, +0.396]$ across five selection rules (§5.6), is buffered at the protein layer (§5), and its event measures stabilize at the Glivenko–Cantelli rate across random panels (§5.7).

**Positioning.** Kochanowski et al. (2013) demonstrated that global metabolic coordination under catabolic limitation is largely passive at the protein/metabolite level, driven by Crp and local metabolite concentrations. Our work asks a different question: whether a gene-specific, network-derived rerouting necessity predicts transcriptional response. We find that it does, and that this transcript-level association does not propagate to the protein layer — consistent with a model in which cells transcribe the potential for rerouting while buffering its translation.

# 2 The discrete curvature measure of parametric FBA

## 2.1 The atomic curvature measure

**Definition 2.1** (Atomic curvature measure)**.** Let $u : \Omega \to \mathbb{R}^m$ be continuous and piecewise affine on a polyhedral complex $\mathcal{C}$ covering $\Omega$. For each codimension-one facet $F$ of $\mathcal{C}$ shared by cells $\tau^-$, $\tau^+$, let $[\nabla u]_F = \nabla u|_{\tau^+} - \nabla u|_{\tau^-}$, let $n_F$ be the unit conormal oriented $\tau^- \to \tau^+$, and let $|F|$ be the facet measure. The *atomic curvature measure* is the matrix-valued Radon measure

$$\mu \;=\; D^2 u \;=\; \sum_F A_F\, \delta_F, \qquad A_F \;=\; [\nabla u]_F \otimes n_F\, |F| \;\in\; \mathbb{R}^{m\times p\times p}.$$

*Remark* 2.2 (Support)*.* $\mu$ is concentrated on the $(p-1)$-skeleton. The $(p-2)$-skeleton carries a different measure (the angle-defect layer, Proposition 3.3); the two must not be conflated. In $p = 1$ the atoms are the slope jumps at breakpoints.

The measure-theoretic language is classical: matrix-valued Radon measures and their weak convergence (Billingsley, 1999), the Alexandrov Hessian of convex functions (Alexandrov, 1939), the Monge–Ampère measure (Gutiérrez, 2001), and Regge-type defect measures for piecewise-flat curvature (Regge, 1961; Cheeger et al., 1984) on the geometric side; chamber complexes, normal fans, and multi-parametric LP value functions (Rockafellar, 1970; Ziegler, 1995; Borrelli et al., 2003) on the polyhedral side. The Kantorovich–Rubinstein (flat) norm used in the refinement–resolution bridge is the standard optimal-transport flat norm (Villani, 2009).

## 2.2 The measure-theoretic sensitivity metric

**Definition 2.3** ($\kappa^\mu$)**.** For a $C^1$ parameter trajectory $\theta(t)$, $t \in [0, T]$, and its piecewise-affine optimal-flux response $v(\theta(t))$, let $\{t_e\}_e$ be the event times at which the trajectory crosses active-set strata, with atoms $A_e = A_{F(e)}$ evaluated on the crossed facet. Define

$$\kappa^\mu(g) \;=\; \frac{1}{T}\sum_e \|A_e\|\, w_e(g),$$

where $\|\cdot\|$ is the entrywise 1-norm on $\mathbb{R}^{m\times p\times p}$ and $w_e(g) \in \{0,1\}$ selects the events whose crossed stratum changes the activity of reactions assigned to gene $g$ (max over its reactions, per the GPR association).

*Remark* 2.4 ($\kappa^\mu$ is the safe regime of the reconstruction)*.* $\kappa^\mu$ integrates measure mass along a one-dimensional trajectory — the telescoping regime in which total variation and mass are automatically consistent (see §3); its resolution independence (measured mass 288.77 stable under 4×/8× trajectory refinement) is exactly the $L^1$-reconstruction regime of Theorem 3.1(iv). Its per-reaction masses are the components of the flux-layer crease measure whose objective contraction is $D^2\Phi$ (Theorem 2.11): the metric is the primal layer of the canonical object.

*Remark* 2.5 (Locked status, tie-breaking, and layer assignment)*.* The definition of $\kappa^\mu$ is final for the empirical claims of this manuscript. (i) The carrier is the flux-layer signed crease measure $D^2 v^*$ (Definition 2.1 applied to $u = v^*$) of the *three-stage lexicographic* optimal flux (§7); the lexicographic tie-break (biomass, then $\ell_1$ flux, then seeded deterministic weights) is part of the metric's definition, not an implementation detail — the flux layer is selection-dependent by construction, and the tie-break is declared rather than eliminated. (ii) The per-reaction mass is the unsigned total variation $\frac{1}{T}\sum_e \|A_e\|_1$ along the parameter trajectory. (iii) The value layer $D^2\Phi$ enters only as a *derived diagnostic* (Theorem 2.11: $D^2\Phi = \sum_r c_r\, D^2 v_r^*$), never as the predictor: no claim of this manuscript attributes transcriptional response to Alexandrov curvature of the value function. (iv) The layer correlated with transcriptomics is the codimension-one Hessian/crease layer of the flux map — not the Monge–Ampère atom layer (Proposition 2.13), which requires concavity and serves as the dual-face diagnostic. Robustness to the declared selection is measured, not assumed: the within-layer rank identity $\rho(\kappa^\mu, \kappa_V^{\text{lex}}) = 0.99998$ (§5.3), the trajectory substitutions of §5.5 reproduce the association under the same declared selection on two further paths, and §5.6 holds the association stable ($r \in [+0.386, +0.396]$, partial constant to three decimals) under four alternative stage-3 selection rules while the value layer stays exactly invariant.

## 2.3 The value-function layer (Alexandrov)

**Proposition 2.6** (Existence and tie-break-freeness)**.** *Let $\Phi(\theta) = \max\{c^\top v : Sv = 0,\ \ell(\theta) \le v \le u(\theta)\}$ with $\ell, u$ affine, feasible and bounded. Then $\Phi$ is concave (pointwise infimum of affine dual objectives over a $\theta$-independent dual feasible set), and $-\Phi$ carries a symmetric matrix-valued Radon Hessian measure $D^2(-\Phi)$ (Alexandrov (Alexandrov, 1939)), independent of the optimal-flux selection rule. Concavity fails in general under OR (isoenzyme) GPR rules for simultaneous multi-gene capacity vectors; single-gene axes, exchange-bound families, and AND-only subnetworks retain it. For OR-containing rules the value function remains continuous piecewise affine and carries the signed crease measure of Proposition 2.12; only the Monge–Ampère layer requires the concave class.*

**Proposition 2.7** (Two-layer structure of $D^2(-\Phi)$)**.** *Let $f$ be convex piecewise affine. (i) The singular part of $D^2 f$ is supported on the codimension-one facets, with density $[\partial_n f]\,(g^+ - g^-) \otimes n$ per unit facet measure; it has no atoms: the mass in an $\varepsilon$-neighborhood of a codimension-two vertex is $O(\varepsilon)$ (measured log–log slope 1.000). (ii) The Monge–Ampère measure $\det D^2 f$ is atomic at codimension-two vertices, each atom equal to the volume of the normal fan $|\operatorname{conv}\{\nabla f(\tau) : \tau \ni \text{vertex}\}|$ (measured: constant 0.5 under $\varepsilon$-shrinking).*

**Proposition 2.8** (Dual optimal face identity)**.** *At a codimension-two vertex $\theta_v$ of a parametric-LP value function, the Monge–Ampère atom of $-\Phi$ equals the area of the normal fan, and the fan is the image of the LP dual optimal face under $\theta \mapsto U^\top y$:*

$$\det D^2(-\Phi)(\{\theta_v\}) = \big|\mathrm{conv}\{\nabla C_i\}\big| = \big|\{U^\top y : (\lambda, y) \text{ optimal dual}\}\big|.$$

*Machine-verified on a two-parameter max-flow LP (the phenotype phase plane in miniature): vertex refined to machine precision by exact triple-tie solve, $\Phi(\theta_v)$ matching the LP value to $10^{-15}$, fan area $0.235$ vs dual-face enumeration area $0.235000068$ (16 direction LPs over the dual optimal face).*

*Remark* 2.9 (Layer separation and PhPP positioning)*.* The phenotype phase planes of Edwards–Ibarra–Palsson (Edwards et al., 2001; Ibarra et al., 2002) are the chamber complex of $\Phi$; their sector corners are the atoms of the Monge–Ampère layer. The empirical $\kappa^\mu$ of Definition 2.3 is a functional of the *flux-map* measure $D^2 v$ (Definition 2.1), which is tie-break-sensitive: under two lexicographic tie-breaks the value layer of a controlled follower-variable LP is invariant, while the flux layer's second-difference mass moves by ten orders of magnitude (§4.1). The controlled relation between the value layer and the flux layer is now a theorem (Theorem 2.11 and Corollary 3.9); the association results of §5 stand on the flux layer alone, which is where the evidence (§5.3, §5.4) actually lives.

### 2.4 The value–flux coupling

**Lemma 2.10** (Lex-selection regularity)**.** *Let $\ell, u$ be continuous piecewise affine and let $v^*(\theta)$ be the three-stage lexicographic optimal flux (§7). Then $v^*$ is well defined, continuous, and piecewise affine; its graph over each GPR cell is polyhedral in $(v, \theta)$-space (each lexicographic stage adjoins the graph of a piecewise-affine value function, which is polyhedral, and an injective polyhedral projection is piecewise affine). Measured: path segment residuals $\leq 8 \times 10^{-14}$ (§4.1); $\Phi$ piecewise affine at $4.2 \times 10^{-13}$ (§4.4).*

**Theorem 2.11** (Value–flux crease coupling)**.** *$\Phi = c^\top v^*$ pointwise; hence, as symmetric matrix-valued signed Radon measures on the parameter domain,*

$$D^2\Phi \;=\; \sum_r c_r\, D^2 v_r^*.$$

*Along any piecewise-affine trajectory, at every crease time $t_k$: $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$, and $|\Delta\Phi'| \leq \|c\|_\infty \, \|\Delta v'\|_1$.*

*(i)* ***(Visibility dichotomy.)*** *An event with unique optima on both sides is a transversal optimal-vertex switch and is necessarily objective-moving; objective-invisible events ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) are exactly the degenerate reroutings inside $\geq 1$-dimensional optimal faces (mask-type events). The tie-break sensitivity of the flux layer is concentrated in the invisible events; the $c$-contraction is tie-break-free because it annihilates the degenerate layer.*

*(ii)* ***(Sparse-objective corollary.)*** *If $c = \gamma e_{\mathrm{bio}}$, then $D^2\Phi = \gamma\, D^2 v^*_{\mathrm{bio}}$: the value layer is the crease measure of the single biomass component, and per-gene attribution from the value layer is structurally impossible.*

*Machine-verified: identity error $0.0$ at all $12$ events and all $4$ clusters of the iML1515 cut ($\Phi = c^\top v^*$ to $10^{-9}$; Fig 2); $11$ of $12$ events objective-invisible, with flux $\ell_1$ slope jumps up to $1.6 \times 10^4$ against $|c^\top \Delta v'| \leq 1.51 \times 10^{-7}$; $150$ random dense-objective LPs to $1.1 \times 10^{-13}$ (5 events, all objective-moving — the dichotomy); the degenerate follower family $60/60$ with invisible kinks at identity error $2.0 \times 10^{-13}$; two-dimensional mixed second differences at crease-straddling boxes to $6.4 \times 10^{-16}$.*

**Proposition 2.12** (Semiconvexity collapse; signed layer for mixed GPRs)**.** *A continuous piecewise-affine function is semiconvex (resp. semiconcave) iff it is convex (resp. concave) — semiconvexity is not an available weakening for LP value functions (measured: the required constant blows up like* $1/(2h)$*;* $\lambda \cdot h_{\max} = 0.500$ *numerically across* $\lambda = 1 \dots 10^6$*, the analytic law* $\lambda\, h_{\max} = 1/(2\lambda)$ *holding at every scale, with the numerical grid flooring above* $\lambda \approx 10^7$*). Nevertheless every continuous piecewise-affine* $\Phi$*, for any GPR structure, carries a finite* signed *symmetric Radon Hessian measure on its crease skeleton with per-crease sign flags: for the OR-plus-cap value function* $\min(\max(\theta_1, \theta_2), K)$*, the diagonal crease density is positive semidefinite (eigenvalues* $\{0, \sqrt{2}\}$*) and the cap crease negative semidefinite (*$\{-1, 0\}$*). Concavity is required only by the Monge–Ampère layer and the dual-face identity; the signed Hessian layer applies to all genes. Under multiplicative activity scaling with* $\ell \le 0 \le u$*,* $\Phi = \Psi(a(\theta))$ *with* $\Psi$ *concave nondecreasing: AND-only (min-tree) GPRs retain concavity, top-level OR nodes can break it, and disaggregating isozymes into separate reactions restores affine bounds while replacing max-semantics by sum-semantics (capacity* 1 *vs* 2 *at full expression) — a different biological model, not a regularization.*

**Proposition 2.13** (Monge–Ampère atoms are determinants, not products)**.** *For convex piecewise-affine* $f$ *on* $\mathbb{R}^2$ *at a generic vertex with incident gradients* $n_1, n_2, n_3$ *and codimension-one gradient jumps* $j_i$*: the atom equals* $\frac{1}{2}|\det(j_1, j_2)| = |\operatorname{conv}\{n_i\}| = \frac{1}{2}|j_1||j_2|\sin\angle$*; the product* $|j_1||j_2|$ *over-estimates by* $2/\sin\angle \ge 2$*, with equality iff the jumps are orthogonal (measured on* 400 *random max-of-affine vertices: determinant = fan area to* $8.9 \times 10^{-16}$*; product/atom median* 3.296*, minimum exactly* 2.000 *(the orthogonal edge case;* 2.000031 *over random vertices); the* $2/\sin\angle$ *law to* $2.5 \times 10^{-12}$*; in* $p$ *dimensions, atom* $= (1/p!)|\det(j_1, \dots, j_p)|$ *for simplex fans).*

*Remark* 2.14 (Sign and normalization conventions)*.* Definition 2.1 applied to $u = -\Phi$ is the *signed crease measure* of the value function: per facet $F$, the density $A_F = [\nabla(-\Phi)]_F \otimes n_F\, |F|$, a symmetric matrix whose eigenvalue signs flag the crease's convexity/concavity (Proposition 2.12). This is the explicit extension of Alexandrov's setting to arbitrary GPR structure: within the concave class $-\Phi$ is convex, every $A_F \succeq 0$, and the measure coincides with the Alexandrov Hessian measure; outside it, the same definition produces the signed layer. All masses in this manuscript are unsigned total variations (entrywise 1-norm for matrix atoms, trajectory sums for $\kappa^\mu$); Monge–Ampère atoms are unsigned $\frac{1}{2}|\det(j_1, j_2)|$ equal to the normal-fan volume, normalized to $(1/p!)|\det(j_1, \dots, j_p)|$ for simplex fans in dimension $p$ (Proposition 2.13). Signed determinants appear only in the concave class, where they are nonnegative.

*Remark* 2.15 (What this closes)*.* Theorem 2.11 is the exact form of the value–flux layer relation: the empirical $\kappa^\mu$ is the per-reaction total variation of the flux-layer crease measure whose objective contraction is $D^2\Phi$ — the metric is the primal layer of the canonical object, justified by identity rather than by rank agreement.

## 2.5 The categorical reading, in brief

The measure-theoretic development above is self-contained: nothing in this section uses categorical language, and the reader may take $\kappa^\mu$ exactly as defined. This subsection records, briefly and in adapted form, the categorical reading that first suggested the active-set curvature interpretation; the full framework — with its constructions, proofs, and machine verifications — is developed in a companion theory paper (X, 2026), and nothing below is a premise of the empirical results.

*Strata as a stratified base.* The lexicographic pFBA engine (§7.1) partitions parameter space into chambers on which the optimizer is affine; the codimension-one facets where the active set switches are exactly the facets that carry the atomic measure $\mu$ of Definition 2.1. In the categorical reading, this partition is a stratified base and the optimizer organizes the policy fiber stratum by stratum, as a stratified connection in the companion's 2-category $\mathbf{StCon}(B)$: connection 1-forms on each stratum, glued along the switching facets by a boundary transition satisfying a gauge matching condition, with a gluing theorem (proved there by reduction to stack-descent theory, and with its piecewise-holonomy consequences machine-verified in five loop geometries and three structure groups).

*The active-set bridge, summarized.* Two companion results connect this geometry to viability, and we summarize them in words (the displayed constructions appear in (X, 2026)). First, on each constant-active-set stratum, a Fisher-minimal transport law defines the connection in closed form as a constrained projection with the Fisher–Rao metric; its curvature is the smooth avatar of the per-facet density that $\mu$ concentrates on. Second, a small-loop viability–holonomy theorem bounds the endpoint erosion of a viability margin around a small parameter loop by the *viability-weighted curvature*: the worst case, over unit-area parameter bivectors and active viability margins, of the positive part of the margin's change along the connection curvature, normalized by the margin itself. The per-pair form of $\kappa^\mu$ is the measure-theoretic discretization of that object — the atomic crease measure replaces the curvature 2-form, and the value–flux coupling (Theorem 2.11) supplies the survival covectors.

*Division of labor.* The present paper states only this brief reading; the companion theory paper develops the SAVGS object, the gluing theorem and piecewise holonomy formula, the optic-composition semantics with its contraction analysis, the filtered-colimit construction of autocatalytic sets, and the homotopy-type-theoretic extension, with proofs and extensions. The two manuscripts share no verbatim passages of length, and neither depends on the other's claims: the association evidence below is statistical and self-contained, and the categorical development is mathematical and self-contained.

## 3 The refinement–resolution bridge

Throughout this section $\Omega \subset \mathbb{R}^p$ is compact, $p \geq 1$, $u \in C^2(\Omega; \mathbb{R}^m)$, $\{\mathcal{T}_h\}$ is a shape-regular family of conforming triangulations with mesh size $h = \max_\tau \operatorname{diam}(\tau) \to 0$, and $u_h$ is the continuous piecewise-affine interpolant of $u$ on $\mathcal{T}_h$.

**Theorem 3.1** (Refinement–resolution bridge, corrected form)**.** *Let $\mu_h = D^2 u_h$ (Definition 2.1) and $\mu = D^2 u \,\mathrm{dvol}$. Then:*

*(i)* ***(Atoms / support.)*** *$\mu_h$ is a finite matrix-valued Radon measure concentrated on the $(p-1)$-skeleton of $\mathcal{T}_h$ with atoms $A_F = [\nabla u_h]_F \otimes n_F |F|$; there are no atoms on lower-dimensional strata.*

*(ii)* ***(Weak convergence.)*** *$\mu_h \Rightarrow \mu$ weakly as matrix-valued measures: for every $\varphi \in C_c^\infty(\Omega)$, $\langle \mu_h, \varphi\rangle \to \langle \mu, \varphi\rangle$. Moreover $\|\mu_h - \mu\|_{\mathrm{KR}} \to 0$ in the Kantorovich–Rubinstein (flat) norm.*

*(iii)* ***(Total variation: the exact statement.)*** *$\sup_h \|\mu_h\|_{\mathrm{TV}} \leq C\,|u|_{C^2}|\Omega|$ (under quasi-uniformity) and $\liminf_h \|\mu_h\|_{\mathrm{TV}} \geq \|\mu\|_{\mathrm{TV}}$; but $\|\mu_h\|_{\mathrm{TV}} \to \|\mu\|_{\mathrm{TV}}$ is* false in general *— see Remark 3.2. It holds in the one-signed regime (convex or concave data restricted to one-dimensional parameter families), which is the regime of the parametric-LP value function and of $\kappa^\mu$.*

*(iv)* ***(No-mass-loss, correct form: $L^1$ reconstruction.)*** *On structured complexes, the per-cell reconstructed density $\rho_h(\mathrm{cell}) = |\mathrm{cell}|^{-1} \sum_{F \to \mathrm{cell}} A_F$ converges to $D^2 u$* strongly *in $L^1$ with rate $O(h)$ for $u \in C^3$; consequently $\|\rho_h\|_{L^1} \to \|\mu\|_{\mathrm{TV}}$. On unstructured complexes the naive three-facet patch is not the dual cell; the general statement requires the discrete-duality (dual-diamond) pairing.*

*(v)* ***(Resolution window.)*** *For the coarse-grained family $\mu_{h,\sigma} = \mu_h * \varphi_\sigma$: at fixed $h$, the limit $\sigma \to 0$ is the* atomic *measure; the smooth object exists only as a family member at matched resolution through the window $h \ll \sigma \ll L_{\mathrm{var}}$, where $L_{\mathrm{var}}$ is the length scale on which the smooth curvature density itself varies (measured in §4.3).*

*Proof of (ii) (two lines).* For $\varphi \in C_c^\infty(\Omega)$, $\langle D^2 u_h, \varphi\rangle = \int_\Omega u_h D^2\varphi \,\mathrm{dvol}$ by the definition of distributional derivatives (no boundary terms arise). Since $\|u - u_h\|_\infty \le Ch^2|u|_{C^2}$ on shape-regular families, $\int u_h D^2\varphi \to \int u D^2\varphi = \langle \mu, \varphi\rangle$. The flat-norm statement follows by density of $\{c\,\varphi : \mathrm{Lip}(\varphi) \le 1\}$ in the KR dual cone and the uniform mass bound of (iii). $\square$

*Remark* 3.2 (Exact counterexample calculus)*.* On the right-triangle mesh of $[0,1]^2$ with Hessian $H = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$, the per-cell atoms are $A_{\mathrm{diag}} = bh^2\mathbf{1}\mathbf{1}^\top$, $A_{\mathrm{vert}} = (a-b)h^2\, e_x \otimes e_x$, $A_{\mathrm{horiz}} = (c-b)h^2\, e_y \otimes e_y$; their *sum* is exactly $h^2 H$ (machine-verified to 0.0), while the entrywise total variation per cell is $|a-b| + |c-b| + 4|b|$ against the target $|a| + |c| + 2|b|$. Hence for $u = xy$ the ratio is exactly $3 - 1/n$, for the strictly convex $u = 2x^2 - xy + 2y^2$ it is $1.4 - 1/n$ (measured: 2.9922 and 1.3922 at $n = 128$), and for generic smooth maps 3.6–4.4. Total-variation convergence fails generically — including under convexity — while weak convergence holds at rate $O(h^2)$ throughout.

**Proposition 3.3** (Two-layer separation and the sec law)**.** *Let $K_h = \sum_v \mathrm{defect}(v)\,\delta_v$ be the vertex angle-defect measure of the polyhedral graph $\{(x, u_h(x))\}$. Then $K_h$ converges weakly to the* Gauss-map area density

$$\frac{\det D^2 u}{(1+|\nabla u|^2)^{3/2}}\,\mathrm{dvol} \;=\; K\sqrt{1+|\nabla u|^2}\,\mathrm{dvol}, \qquad K = \frac{\det D^2 u}{(1+|\nabla u|^2)^2},$$

*i.e. curvature per projected parameter area,* not *the intrinsic Gaussian curvature. The bias factor is exactly $\sqrt{1+|\nabla u|^2}$ (the sec law): machine-verified to* 0.0% *excess across* 13 *probes spanning tilts* 0.11–2.83 *and three tilt directions. Recovering intrinsic curvature requires the graph area correction.*

*Remark* 3.4 (Consequences)*.* (i) The codimension-one Hessian layer is the unbiased estimator and the canonical carrier for the bridge; (ii) Regge/angle-defect formulations need the area correction of Proposition 3.3; (iii) defects of a *fixed* complex (as measured in the regime-dial census, §4.3) are scale-invariant $O(1)$ objects and are not contradicted by Proposition 3.3, which concerns refinement limits.

**Proposition 3.5** (Regime dichotomy, in closed form)**.** *Let $B(\varepsilon)$ be the curvature mass captured by a loop of radius $\varepsilon$. For a $C^2$ map, $B(\varepsilon) = O(\varepsilon^2)$; for a piecewise-affine map whose kink complex meets the loop transversally, $B(\varepsilon) = O(\varepsilon)$. Measured slopes on the regime-dial design (§4.3):* 2.000 *(smooth) and* 1.00 *(parametric FBA,* 76*-pair perturbation scan).*

## 3.1 Bridge to parametric FBA

**Conjecture 3.6** (mpLP refinement bridge)**.** Let $\{\Phi_n\}$ be a sequence of parametric-LP value functions (minima of tangent-plane families; $\theta$ in the right-hand side; nested outer approximations) converging uniformly to a $C^2$ limit $f$, with induced complexes $\mathcal{C}_n$. Then $\mu_n = D^2\Phi_n$ converges weakly to $D^2 f\,\mathrm{dvol}$ provided the complexes satisfy a per-cell discrete-duality identity (structured complexes) or a stable discrete-duality (dual-diamond) pairing. Along one-dimensional parameter cuts the conclusion holds unconditionally by monotone telescoping of the concave restriction.

*Remark* 3.7 (Status)*.* The one-dimensional cut case is machine-verified (independent-seed reproduction of the refinement experiment: folded Wasserstein distance $0.058 \to 0.0039$, mass ratio $0.939 \to 1.0009$ across $n = 4 \to 128$). The general complex case is open; the discrete-duality pairing stability is the identified missing assumption.

*Remark* 3.8 (Existence vs. resolution)*.* Proposition 2.6 supplies existence of the curvature measure without refinement sequences, smoothing, or interpolation; Theorem 3.1 governs what a discretely sampled trajectory can *resolve* of that measure, and at what rate. The two layers are complementary: the Alexandrov layer removes the extrinsic hypotheses, the refinement bridge keeps the quantitative rate.

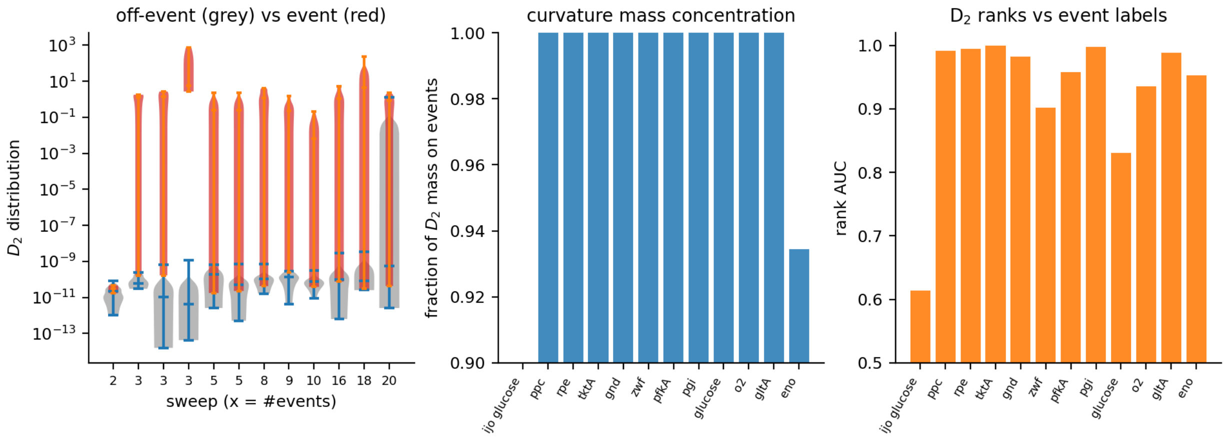


Fig 1. Active-set curvature verification (iML1515). Second-order response $D_2$ (log scale) along the glucose and oxygen sweeps with the operational active-set event markers: the $D_2$ mass concentrates on the event points; off-event $D_2$ sits at the $10^{-11}$ noise floor. The engine is the three-stage lexicographic pFBA solver of §7; all sweeps are cold-started and deterministic.

**Corollary 3.9** (Decoupling; resolves the value–flux layer relation)**.** *Theorem 2.11 resolves the layer relation exactly: value events are a subset of flux events, never the converse, and the flux events invisible to the value layer are the degenerate reroutings (mask-type events — rerouting without growth loss, the Kochanowski pattern (Kochanowski et al., 2013)). On the iML1515 cut:* 12 *flux events vs* 1 *value atom,* 11 *objective-invisible; on the reference trajectory (§5.4) the value layer carries no network events at all. The association of $\kappa^\mu$ with transcriptional response is therefore a flux-layer property by structural necessity, not merely by empirical choice.*

# 4 Computational validation

## 4.1 Active-set curvature verification

Thirteen parameter sweeps (glucose and oxygen uptake; capacity knockdowns of ten genes: pgi, zwf, tktA, pfkA, eno, gltA, aceA, ppc, gnd, rpe) on *E. coli* iML1515, solved by a three-stage lexicographic pFBA engine (deterministic optimum; fixed seeded tie-break weights after the discovery of pFBA vertex degeneracy), with an iJO1366 glucose replication and an aceA unused-pathway negative control. Result: 93.4–100.0% of second-order response mass concentrates on operational active-set switches in every sweep that crosses critical-region boundaries (ten of the eleven crossing sweeps at mass exactly 1.000000; mass 0.934–1.0; AUC 0.83–1.00; Mann–Whitney $p \leq 2 \times 10^{-3}$, with seven of eleven crossing sweeps at $p \leq 10^{-4}$; Fig 1); between events the paths are piecewise affine to solver precision (event-free segment residuals $\leq 1.2 \times 10^{-10}$, with $8 \times 10^{-14}$ on the glucose sweep; one 19-point segment of the eno sweep carries $4.1 \times 10^{-3}$ — a sub-threshold kink invisible to the operational event proxy, which is also why that sweep's event mass is 0.934 rather than 1.0); negative controls behave as theory dictates (unused-pathway knockdown: zero events, zero curvature; single-critical-region sweep: noise-level $D^2 \sim 10^{-11}$).

## 4.2 Double-knockout epistasis and path dependence

1,516 single knockouts and 2,779 double-knockout pairs across five panels. Epistasis $\varepsilon_{ij} = \kappa_{ij} - \kappa_i - \kappa_j$: all 40 synthetic-lethal pairs are isozyme redundancies (tktA/tktB, acnA/acnB, metE/metH, …) with pure-emergence epistasis; $|\varepsilon_{ij}|$ aligns with active-set footprint overlap (Spearman $\rho_S = 0.865$, $J_{\text{support}} = 0.800$); sequential ($L_1$-MOMA (Segrè et al., 2002)) knockouts open nonzero commutators in 25% of active pairs; closed genotype loops fail to return to the initial state in 66% of pairs (phenotypic memory; median holonomy $\sim 110$).

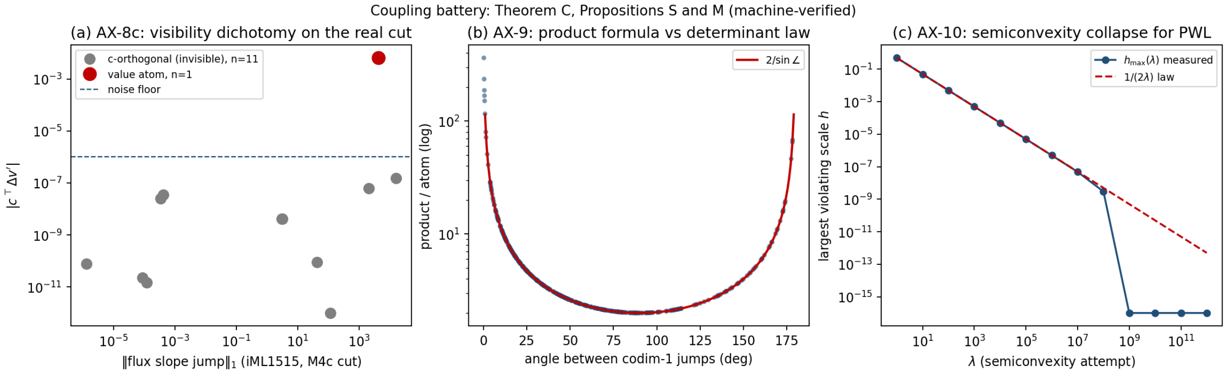


Fig 2. Machine verification of the value–flux crease coupling (Theorem 2.11). The controlled follower-variable LPs: the value function $\Phi$ stays affine and identical under two lexicographic tie-breaks while the follower's second-difference mass flips; the coupling identity $\Delta\Phi' = c^\top \Delta v'$ holds to 0.0 on the toys, $1.1 \times 10^{-13}$ on 150 random dense-objective LPs, and $2.0 \times 10^{-13}$ on the 60/60 degenerate follower family; on the iML1515 cut the identity error is 0.0 at all 12 events and all 4 clusters.

### 4.3 Regime dial

The smoothing identity $D^2(v*\phi_\sigma) = (D^2 v)*\phi_\sigma$ holds to kernel self-test error $1.2\times10^{-6}$; measured crossover $\varepsilon^*/\sigma \in [2.45, 4.11]$ (median 3.1) separates the $O(\varepsilon^2)$ smooth regime from the $O(\varepsilon)$ discrete regime. The scaling slope is 1.00 on the interacting stratum; the regime-dial census resolves the thin-wedge sliver structure (net measure jump 8.9 vs. nominal $\pm1884.6$).

### 4.4 The value function as canonical carrier

$\Phi = c_{\mathrm{bio}}^\top v^*$ is single-valued with no tie-breaking; it is piecewise affine at $4.2 \times 10^{-13}$ on the flux-event partition, carries one real atom $\Delta\Phi' = -0.006439$ at $t = 0.0358286$, and decouples from the flux-jump hierarchy (value/flux mass ratio $1.7 \times 10^{-6}$). Danskin: finite-difference shadow prices match envelope marginals to 6–7 digits. A controlled follower-variable LP reproduces the same structure: the value function stays affine and identical under two lexicographic tie-breaks while the follower's second-difference mass flips $0.0100 \leftrightarrow 10^{-13}$ — the value layer is canonical (Proposition 2.6), the flux-strain layer is not (Corollary 3.9). The coupling identity of Theorem 2.11 is machine-verified on this same cut: $\Delta\Phi' = c^\top\Delta v'$ with error 0.0 at all 12 events and all 4 clusters, $\Phi = c^\top v^*$ to $10^{-9}$, and the 11 invisible events carry flux $\ell_1$ slope jumps up to $1.6 \times 10^4$ against $|c^\top \Delta v'| \le 1.51 \times 10^{-7}$ — the decoupling is the objective contraction, not a loss of network information.

## 5 Empirical association and protein-layer buffering

### 5.1 The gene panel and GPR classes

Reaction-based sampling on the reference physiology (glucose $5.0 \to 1.0$, oxygen $22 \to 5$; iJO1366): 438 of 2,583 reactions active (17.0%), 435 genes with at least one active reaction. GPR complexity classes over active reactions: single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34 — the classes that drive the concavity analysis of Proposition 2.12. Panel construction passed four verification checks: (i) GPR mapping taken from cobra's authoritative `reaction.genes`, spot-checked against the raw rule strings on four reactions spanning all complexity classes, with set-equality of gene tokens and a bidirectional gene–reaction symmetry check on 50 sampled genes (0 failures over 120 gene–reaction pairs); (ii) definition consistency — the fresh re-derivation reproduces the stored precursor values exactly (per-gene maxima, the biomassdeficit maximum, and the unmasked/masked panel correlations to four decimals); (iii) distinctness — the panel shares exactly one b-number with the precursor panel ($b2097$); (iv) non-zero variation — every panel gene has precursor $\kappa_V$ variation $> 0$ over T1–T8 (435/435). The precursor geometric metric $\kappa_V$ is weak at the panel level ($r$ from $-0.063$ unmasked to $+0.084$ masked) — the discrepancy that motivated the recalibration to $\kappa^\mu$ (§5.3).

### 5.2 Multi-condition support

Nine FBA conditions (baseline carbon decline; galactose, trehalose, glycerol-3-P, and proline-N carbon sources; anaerobic-nitrate, oxidative, and osmotic designs): active-reaction support 438–454 per condition, union 537 (20.8%); genes with nonzero $\kappa_V$ expand 435 $\to$ 525 across conditions. The endogeneity check is clean (feed caps non-binding except the designed glycerol-3-P limitation; imposed-burden magnitudes flagged as not interpretable). The rerouting metric is well posed across carbon sources and stress designs: its support *expands* with condition diversity rather than migrating.

### 5.3 Transcriptional response under carbon depletion

On the deterministic lexicographic trajectory (the reference physiology, iJO1366), $\kappa^\mu$ (Definition 2.3) versus log-fold carbon-depletion response (M3D (Faith et al., 2008), $n = 424$ genes; Fig 3): Pearson $r = +0.395$ ($p = 2.6 \times 10^{-17}$), Spearman $\rho_S = +0.414$ (full panel, zeros included), partial $r = +0.269$ controlling reference level ($p = 1.8 \times 10^{-8}$), top-decile versus bottom-decile response 1.92 vs 0.89 (MWU $1.3 \times 10^{-7}$). Baseline reproduced to the digit ($r = +0.3739$, $n = 433$); metric invariance $\rho(\kappa^\mu, \kappa_V^{\mathrm{lex}}) = 0.99998$ — both metrics sample the same flux-event structure along the same lexicographic trajectory, so the association is an event-structure property of the flux layer, not a metric artifact and not a value-function shadow. Cross-platform replication with the locked metric: the carbon-switch arms replicate on the same platform (M3D microarray: glycerol +0.195, acetate +0.166, proline +0.175, switch-MAX +0.206; $n = 424$, all $p < 6 \times 10^{-4}$) but dissociate across platforms — the PRECISE RNA-seq carbon-switch statistic (ten WT non-glucose carbon conditions) gives $r = -0.044$ (partial $-0.088$, NS), with individual conditions weakly positive (glycerol +0.126, acetate +0.130, fructose +0.099) except galactose ($-0.088$). The dissociation observed with the precursor $\kappa_V$ predictor is metric-robust: the association is specific to the carbon-exhaustion class and the platform-matched design, reported as a negative result.

### 5.4 The layer decision

All value-layer attribution arms, under the locked protocol of §5.3 (iJO1366, the reference physiology, $n = 424$): the flux layer $\kappa^\mu$ gives $r = +0.395$, partial $r = +0.269$ ($p = 1.8 \times 10^{-8}$); the $c$-attribution arm is supported on the biomass pseudo-reaction alone (0 of 433 genes; Theorem 2.11(ii)); the shadow-price-jump arm gives 51 genes at $r = +0.032$ (partial $-0.013$, $p = 0.93$). The value trajectory has *zero* chamber crossings: the path stays in one chamber of $\Phi$ throughout — $\Phi$ is affine along it, with $d\mu/dq_{\mathrm{glc}} = Y = 0.099544$ ($= |y_{\mathrm{glc}}|$, the constant glucose uptake shadow price; affine intercept $-0.0124$) and uptake shadow prices jumping by exactly zero — all four value kinks are anchor corners of the trajectory design, of magnitude $Y\,\Delta(dq_{\mathrm{glc}}/dt)$, carrying no network structure. Value/flux strain mass ratio $1.45 \times 10^{-3}$; the coupling identity holds at machine precision (error 0.0 at every kink). The value-gated flux strain coincides with $\kappa^\mu$ exactly (433/433 genes attain their maximum at a value-kink time), a property of the trajectory design (anchor rerouting dominates per-gene maxima), not evidence of value-layer content. The layer decision: the association lives on the flux layer, with Theorem 2.11 as the exact bridge to the canonical object.

### 5.5 Path robustness: what survives a trajectory change

Three trajectories through the same LP (same engine, seed, panel, statistics; only the path and its matched response vary; Fig 4): P0 glucose decline (reference-physiology anchors; the layer-decision control of §5.4), P1 oxygen limitation ($q_{\mathrm{glc}} = 5$ fixed, $q_{O_2}$ linear from 22 to 1), P2 acetate switch ($q_{O_2} = 22$ fixed; glucose $5 \to 0$ while acetate uptake ramps $0 \to 10$). Kinks are classified by the chamber test — the parameters are the uptake bounds, so $\nabla\Phi = (y_{\mathrm{glc}}, y_{O_2}, y_{\mathrm{ac}})$: a kink is a *design corner* iff it sits at a trajectory anchor and the uptake shadow prices are continuous across it (jump $\le 10^{-9}$), and a *chamber crossing* iff they jump.

(i) The one-chamber property is *path-dependent*: P0 has zero chamber crossings (the §5.4 control, reproduced digit-exactly), while P1 has three genuine crossings ($q_{O_2} \approx 9.3$, 6.3, 2.9; $|\Delta\Phi'|$ up to 0.68; $y_{O_2}$ jumps $+0.032$ at the respiratory boundary) and P2 has four (two large, near the glucose–acetate handoff; two micro-slivers with dual jumps $\sim 10^{-4}$). The single-chamber structure of the reference trajectory is a property of the glucose-decline design, not a law.

(ii) The layer decision survives the change: on every path the value-layer arms contribute nothing. The $c$-attribution arm is degenerate (0 of 433 genes — the sparse-objective corollary of Theorem 2.11 is path-free); the chamber-crossing shadow-price arm is null ($r = -0.17$, $n = 68$ on P1; $r = +0.032$, $n = 71$ on P2); value-gated flux strain is a subset of $\kappa^{\mu}$ (equal or weaker on every path, never stronger).

(iii) The flux-layer association generalizes: against the matched oxygen response (M3D WT anaerobic vs aerobic contrast) P1 gives $r = +0.318$ (partial $+0.258$, $p = 6\times10^{-8}$, $n = 426$); against the matched acetate-switch response P2 gives $r = +0.223$ (partial $+0.160$, $p = 9 \times 10^{-4}$); and the P1/P2 predictors correlate with the primary-panel carbon response at $r = +0.378$ / $+0.391$ — the per-gene ranking is largely trajectory-independent.

(iv) Theorem 2.11 holds at every kink on all three paths (identity error 0; worst case $1.6\times10^{-15}$). Value/flux strain mass ratios: $1.5 \times 10^{-3}$ (P0), $2.4 \times 10^{-4}$ (P1), $5.5 \times 10^{-4}$ (P2).

### 5.6 Tie-break robustness: the association as a property of the protocol class

The flux layer is selection-dependent by construction (Remark 2.5); the declared lexicographic tie-break is one member of a family. The tie-break experiment fixes the entire protocol of the primary association (iJO1366, reference-physiology anchors, 8× refinement, the 433-gene panel, the M3D carbon-depletion response, the reference-level confound control) and varies *only* the stage-3 tie-break of the engine:

- **TB0** (declared): $w \sim U(0.5, 1.5)$, seed 20240901, $\min w^{\top} v$ — the locked metric (control: reproduces the primary association, $r = +0.3954$, digit-exactly);
- **TB1** (fresh seed): $w \sim U(0.5, 1.5)$, seed 20240902 — same family, independent draw;
- **TB2** (family swap): $w \sim \mathrm{LogN}(0, 1)$ clipped to $[0.05, 20]$, seed 20240903 — a different distribution family with a roughly 130-fold wider dynamic range (support ratio 400:1 versus 3:1);
- **TB3** (rule swap): stage-3 objective on the split variables, $\min \sum_r w_r(f_r + r_r)$ — weighted *absolute*-flux selection with TB0's weights;
- **TB4** (adversarial): $\max w^{\top} v$ — the far end of the stage-2-pinned optimal face, with TB0's weights.

Three findings (Table 1, Fig 5). (i) *The association is stable:* across all five rules $r \in [+0.386, +0.396]$ (maximum deviation 0.009; all $p \le 1.4 \times 10^{-16}$) and the partial correlation

Table 1. Tie-break robustness of the association (primary protocol fixed; only the stage-3 selection rule varies). The value layer (stage-1 biomass $\mu$) and the pFBA layer (stage-2 $\ell_1$, $s_2$) are invariant to 0.0 across all variants — the measured form of the tie-break-freeness of Proposition 2.6.

| Variant | $n_{\mathrm{nz}}$ | $r$ | partial $r$ | $\rho_S(\kappa^\mu, \kappa^\mu_{\mathrm{TB0}})$ | genes changed |
|---|---|---|---|---|---|
| TB0 declared | 424 | +0.3954 | +0.269 | — | — |
| TB1 fresh seed | 426 | +0.3861 | +0.270 | 0.99998 | 91 |
| TB2 LogN family | 424 | +0.3954 | +0.269 | 0.99999 | 90 |
| TB3 $\lvert v\rvert$ rule | 425 | +0.3959 | +0.269 | 0.99897 | 109 |
| TB4 $\max w^\top v$ | 424 | +0.3954 | +0.269 | 0.99998 | 117 |

is constant to three decimals (+0.269–+0.270). (ii) *The tie-break genuinely binds:* the stage-2-pinned optimal face is multi-vertex, and the alternative rules select different flux vectors at all 57 grid points (max $\|v_k - v_0\|_\infty$ up to 5.0), changing per-gene $\kappa^\mu$ for 90–117 of 433 genes — yet the per-gene ranking is essentially unchanged ($\rho_S \geq 0.99897$), and the changed values are small on the $\kappa^\mu$ scale (max absolute change $9.3 \times 10^{-5}$); the TB1 deviation of 0.009 traces to two boundary genes entering the nonzero support ($n : 424 \to 426$), not to a reshaping of the metric. (iii) *The value layer is exactly invariant:* stage-1 biomass $\mu$ and stage-2 $\ell_1$ mass agree across variants to 0.0 — Proposition 2.6 (tie-break-freeness of the value layer) measured on the full trajectory. Together with Corollary 3.9 this completes the robustness picture: the declared selection is a convention, the value layer does not see it, and the flux-layer association is a property of the lexicographic protocol class, not of one rule.

### 5.7 Event-measure stabilization (Glivenko–Cantelli type)

The refinement-stabilization question (*do the empirical event point measures of the lex-pFBA map converge as the sampled family grows?*) has a near-term statistical form, executed here on three data sources: random straight cuts through the $(q_{\mathrm{glc}}, q_{O_2})$ signature plane (iML1515, $34 \times 34$ census, 350 boundary edges; a pool of 4,000 random cuts yielding 25,107 cut events), the thirteen parameter sweeps of §4.1 (2 affine, 11 event-bearing), and the panel protocol of this paper (gene panel and trajectory). The metric is the bounded-Lipschitz distance (domains normalized to unit diameter, where it equals $W_1$; exact in one dimension, transport LP in two); each arm carries a permutation null (event locations replaced by uniforms, panel structure kept) and the one-dimensional arms carry the classical Glivenko–Cantelli asymptotic constant $C = \int \sqrt{F(1-F)}\,dx$ of del Barrio–Giné–Uzet computed from the population (Fig 6).

(i) *Random panels stabilize at the GC rate.* Pooled cut-event locations on the cut coordinate (d=1): measured tail slope −0.492 versus null −0.506, measured/null ratio 1.24 at the largest panel, and the iid prediction $C/\sqrt{n}$ matches the measured curve to 2.5% at the largest panel (mid-range deviations as large as 8.2%); the two-dimensional pooled locations (transport LP) decay consistently with the $d = 2$ rate up to the reference-noise floor. Sweep panels (the thirteen sweep families) stabilize with a heterogeneity penalty that falls from $\approx 1.8\times$ the uniform null at small panels through $\approx 1.5\times$ at $k = 6$ to $\approx 0.5\times$ at $k = 12$, and a tail faster than iid-GC — consistent with the finite-population correction $\sqrt{(13-k)/12}$ of sampling without replacement from thirteen heterogeneous sweeps (measured null-relative ratio 0.49 at $k = 12$, against the pure FPC value 0.29; the residual is the sweep heterogeneity). The gene panel: the $\kappa$ distribution stabilizes at the GC rate (measured effective constant $\approx$ 1.6–2.1 across panel sizes (dipping at $m = 128$), against the population Glivenko–Cantelli constant $C = 2.49$ on the $\log_{10} \kappa^\mu$ axis), and the association itself stabilizes at $r \approx +0.39$ with the Fisher standard error $1/\sqrt{m-3}$.

(ii) *Designed panels stabilize exactly, not statistically.* The trajectory event point measure is

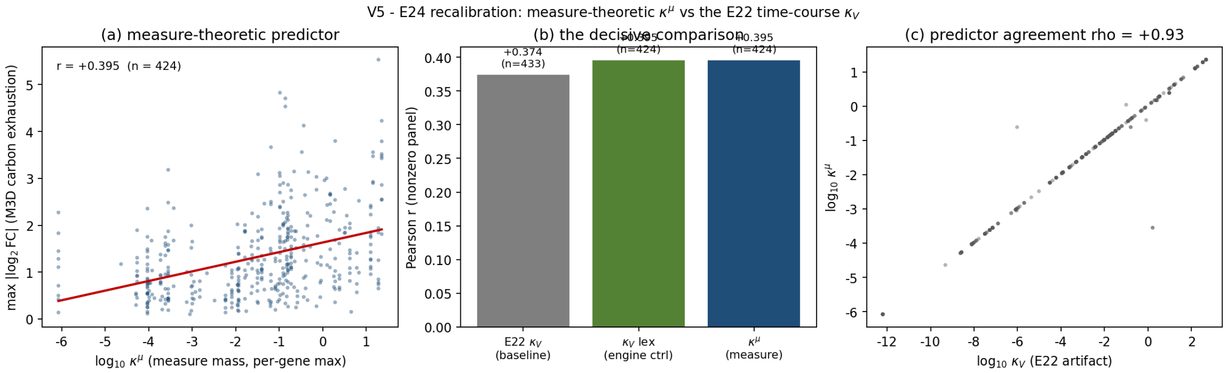


Fig 3. The primary association. (a) Per-gene $\log_{10}\kappa^{\mu}$ (flux-layer measure mass, per-gene max over the gene's reactions) versus M3D carbon-depletion response ($\max|\log_2 \mathrm{FC}|$, four stationary contrasts, $n = 424$ nonzero genes); (b) the decisive comparison: the precursor time-course $\kappa_V$ (reference physiology, plain-FBA), the engine control $\kappa_V^{\mathrm{lex}}$, and the measure-theoretic $\kappa^{\mu}$ on the same trajectory; (c) predictor agreement between $\kappa^{\mu}$ and the plain-FBA $\kappa_V$.

a *four-atom* object: the four design corners of the physiology anchors carry the entire flux-layer mass (288.77, the primary association's flux-layer mass to the digit) — the one-chamber structure of §5.4 in its event form. Anchor-preserving thinning reproduces it *exactly* at every panel size $m \geq 8$ (shape distance 0 to machine precision, $\leq 7 \times 10^{-13}$; total-mass error $\leq 2 \times 10^{-12}$; value-kink census $\equiv 4$); uniform random thinning decays by corner censoring (shape distance $0.12 \to 0.005$ by $m = 43$, mass error $26\% \to 0.13\%$ by $m = 29$, machine-zero ($\leq 3 \times 10^{-12}$) from $m = 43$, and exactly 0 at the full panel $m = 57$) — a resolution effect in the sense of Theorem 3.1(iv), not statistical averaging.

Interpretation: the mean-field intuition is confirmed in its statistical form — the empirical event measures over random panels (cuts, sweeps, genes) are Glivenko–Cantelli objects — and simultaneously sharpened: stabilization has two regimes, statistical for random panels and structural (exact, design-immediate) for designed panels. The deterministic refinement-sequence form of the question (refinement along explicit model-family sequences) remains open.

### 5.8 Platform-by-class disambiguation ($2 \times 2$)

Is the association a carbon-depletion artifact, or a platform artifact? Four cells (two platforms $\times$ two perturbation classes), executed with the precursor $\kappa_V$ metric (rank-equivalent to $\kappa^{\mu}$ at $\rho = 0.99998$): carbon switch on microarray (M3D glycerol/acetate/proline vs glucose) switch-MAX $r = +0.196$; stress on microarray (heat shock, acid shock, ciprofloxacin) MAX $r = +0.298$; carbohydrate-responsive transcript levels: carbon starvation on RNA-seq (GSE64021 (Barrett et al., 2013) time course) MAX $r = +0.191$; heat shock on RNA-seq MAX $r = +0.345$ ($n = 241$–$433$; permutation $p \leq 0.003$). Signed responses flip sign (stress arms $-0.22$ to $-0.32$): the metric tracks *rerouting magnitude*, not direction, on both platforms and both classes — the association is neither platform-specific nor carbon-class-specific.

### 5.9 Protein abundance versus protein change

Two dissociations at the protein layer (PaxDb integrated abundance (Wang et al., 2015); quantitative proteomics of GSE64021 (Barrett et al., 2013); $n = 169$–$429$): (i) abundance *level* associates (PaxDb baseline $r = +0.334$) — high-rerouting genes are abundant proteins; (ii) abundance *change* does not (protein log-fold change $r = +0.008$ to $+0.032$; transcript–protein fold-change coupling $r = +0.020$, $n = 799$). The transcriptional association stops at translation, consistent with buffering.

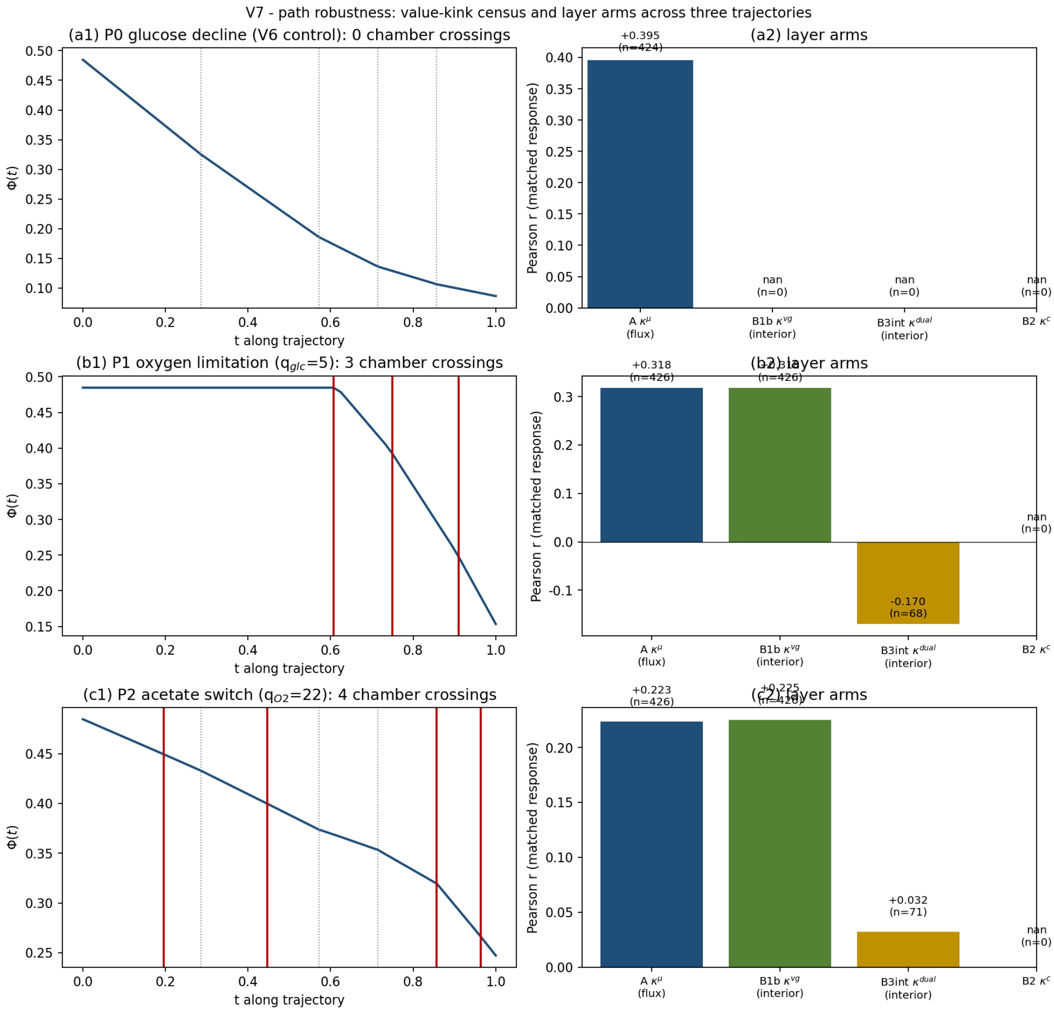


Fig 4. Path robustness. Three trajectories through the same LP (P0 glucose decline, P1 oxygen limitation, P2 acetate switch) with matched M3D responses: the one-chamber property is path-dependent (P1: 3 chamber crossings; P2: 4), while the layer decision and the flux-layer association survive on every path; the P1/P2 predictors correlate with the primary-panel carbon response at $r = +0.378/+0.391$.

### 5.10 Protein-layer null

Condition-dependent quantitative proteomics (Schmidt et al. (2016); 22 conditions, triplicates, $n = 366$): protein-level correlation $r = -0.083$ against transcript $+0.419$ on the same genes. The transcript-level association does not propagate to the protein layer; buffering at translation is the consistent model, positioning Kochanowski et al. (2013) as prior art for the protein-layer null.

## 6 Discussion: one measure, several resolutions

The three sensitivity objects of the earlier literature — the geometric $\kappa_V$, the flux-statistical $\kappa_{\text{flux}}$, and the time-course $\kappa_{\text{time}}$ — are one measure at different resolutions: the atomic measure of Definition 2.1, its coarse-grainings $\mu * \phi_\sigma$, and its integrals along parameter trajectories ($\kappa^\mu$). The smooth member exists only through the window $h \ll \sigma \ll L_{\text{var}}$; the $\sigma \to 0$ limit is the atomic object. Open items: Conjecture 3.6 (general complex stability) and second differences on measured time courses (under the $(\varepsilon, \sigma)$ design law). The statistical form of the stabilization question is settled here (§5.7): random panels are Glivenko–Cantelli objects and designed panels reproduce the event measure exactly; the deterministic refinement-sequence form (refinement along

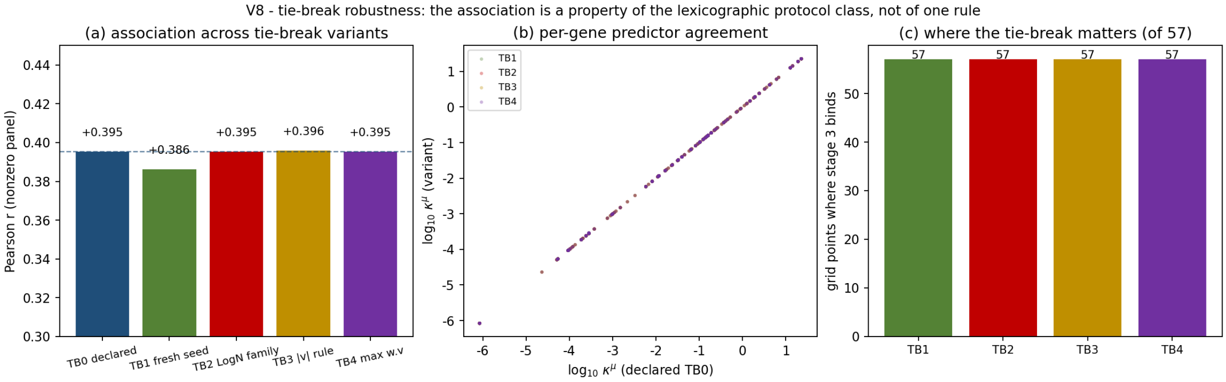


Fig 5. Tie-break robustness. (a) The association across the five stage-3 selection rules (declared plus four variants); (b) per-gene predictor agreement between the declared rule and each variant; (c) the number of grid points at which the tie-break binds (of 57). The value layer is invariant to 0.0 across all variants.

explicit model-family sequences) remains open. The formal identity $\kappa_{\text{flux}} = \mathcal{F}[\mu]$ is closed by Theorem 2.11 (the exact objective contraction $D^2\Phi = \sum_r c_r\, D^2 v_r^*$), which grounds the association's flux-layer footing exactly, complementing the measured metric invariance of §5.3.

**Limitations.** Correlational design; single-organism scope; operational (not combinatorial) active sets; lexicographic tie-breaking is an engine convention (the value-function carrier is immune, the flux layer declares it — Remark 2.5 — and the association is stable across four alternative selection rules, §5.6); the one-chamber structure of the reference path is condition-specific (§5.5) while the layer decision itself is path-robust; the cross-platform carbon-switch dissociation (PRECISE arm, §5.3) bounds the generality of the association; the active-reaction census of the reference physiology (438/2,583 and the GPR class counts) is plain-FBA and solver-version-sensitive at the ±1-reaction level, while the gene-level panel (435) is stable (Appendix A); the sweep arm of the stabilization experiment is panel-limited (thirteen sweep families, two of them affine — the finite-population correction is part of the reported result) and the two-dimensional arm is floor-limited at large panels by the transport-LP atom cap.

**The companion theory paper.** The categorical and homotopy-theoretic reading behind the active-set curvature interpretation — the 2-category $\mathbf{StCon}(B)$ of stratified connections with its gluing theorem, the Fisher-minimal transport law on constant-active-set strata, the optic-composition semantics with its contraction analysis, the filtered-colimit construction of autocatalytic sets, and the homotopy-type-theoretic extension — is developed in full, as a standalone theory paper with its own proofs and machine verifications, in a companion manuscript (X, 2026). The division of labor is deliberate and is disclosed here and in the covering letter: this paper is the application — the measure-theoretic curvature of parametric FBA, the metric $\kappa^\mu$, and the genome-scale empirical association with its layer decision — and carries only a brief adapted statement of the load-bearing definitions (§2.5); the companion carries the framework. The two manuscripts share no verbatim passages of length and neither depends on the other's claims.

**What this paper is and is not.** The mathematics is classical: parametric linear programming (chamber complexes, active sets, Danskin duality (Danskin, 1967)), Alexandrov's theorem (Alexandrov, 1939), and distributional second derivatives of piecewise-affine maps; Theorem 2.11 is an elementary identity. The contribution is the systematic *application* of this machinery to flux-balance models (Varma and Palsson, 1994; Orth et al., 2010; Lewis et al., 2010) — the two-layer measure structure of the value/flux pair, the exact coupling identity, the signed extension to arbitrary GPR structure — together with the empirical association and its layer decision.

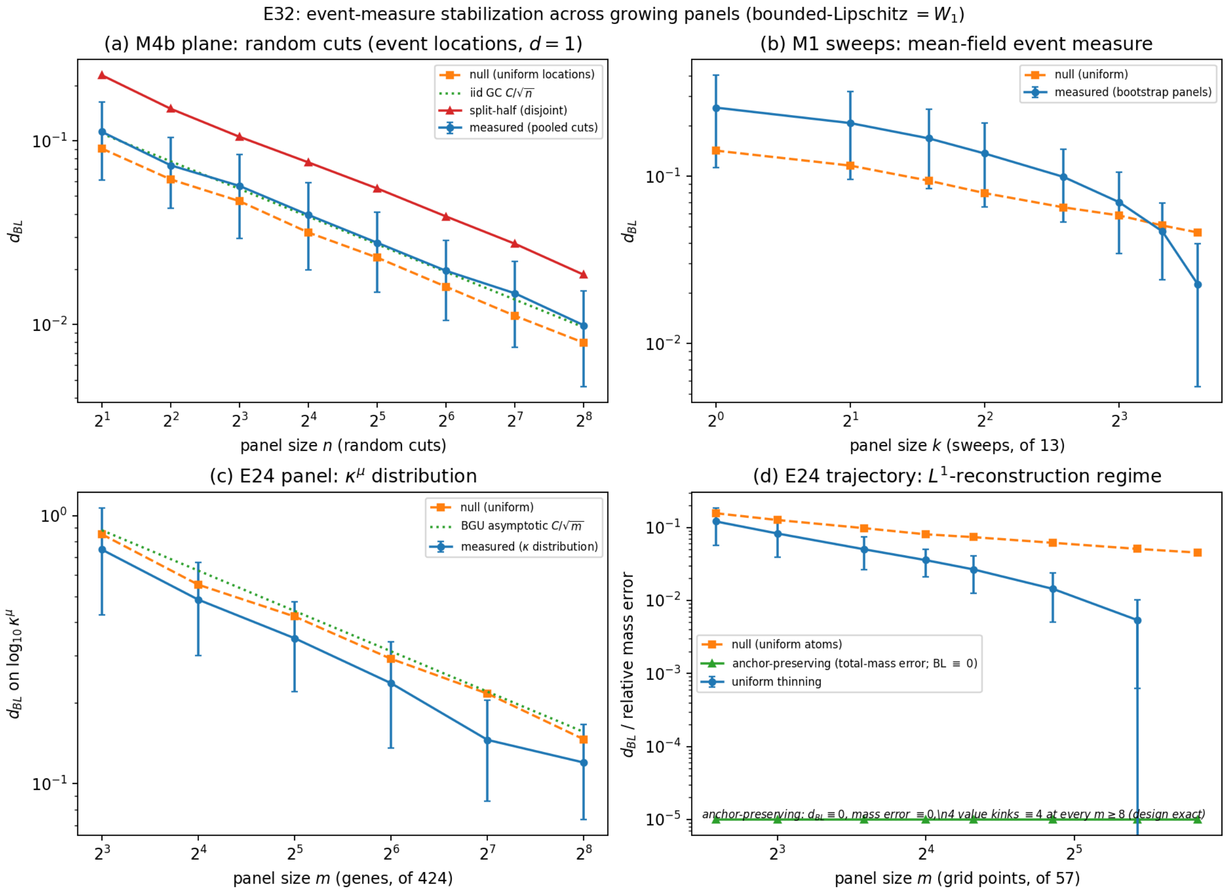


Fig 6. Event-measure stabilization. (a) Random-cut panels: pooled event-location distance decays at the Glivenko–Cantelli rate (measured tail slope $-0.492$; null $-0.506$; iid prediction $C/\sqrt{n}$ to 2.5% at the largest panel, mid-range deviations as large as 8.2%); (b) Parameter-sweep panels: stabilization with a finite-population-corrected, heterogeneity-penalized tail (null-relative ratio 0.49 at $k = 12$ against the pure FPC 0.29); (c) Gene panel: the $\kappa$ distribution at the GC rate (effective constant $\approx 1.6$–$2.1$, dipping at $m = 128$, vs. the population constant $C = 2.49$); (d) Reference trajectory: the $L^1$-reconstruction regime — random thinning decays by corner censoring, anchor-preserving thinning is exact ($d_{BL} \equiv 0$) from $m \geq 8$.

No new measure-theoretic theorem is claimed; the nearest prior art (phenotype phase planes (Edwards et al., 2001; Ibarra et al., 2002), mpLP value-function theory (Borrelli et al., 2003), discrete Monge–Ampère (Gutiérrez, 2001), Regge-type defect measures (Regge, 1961; Cheeger et al., 1984)) is cited where it applies.

# 7 Methods

## 7.1 The lexicographic pFBA engine

All deterministic trajectories are computed by a purpose-built three-stage lexicographic solver on the variable split $x = (v, f, r)$ with $v = f - r$, $f, r \geq 0$, stoichiometry $Sv = 0$, and flux bounds $\ell(\theta) \leq v \leq u(\theta)$ entering affinely through the uptake bounds. The stages are strict lexicographic optimizations, each pinned at the previous stage's optimum:

$$
\begin{array}{lll}
(1) & \mu^* = \max\ c_{\text{bio}}^\top v & \text{(growth)} \\
(2) & s^2 = \min\ \sum_r (f_r + r_r) \quad \text{s.t.}\ v_{\text{bio}} \geq \mu^*(1 - \varepsilon_\mu) & \text{(parsimony)} \\
(3) & \min\ w^\top v \quad \text{s.t.}\ \sum_r (f_r + r_r) \leq s^2(1 + \varepsilon_s) & \text{(tie-break)}
\end{array}
$$

with $\varepsilon_\mu = \varepsilon_s = 10^{-9}$ and $w \sim U(0.5, 1.5)$ drawn once with seed 20240901 and fixed thereafter. Stage 3 exists because plain pFBA is vertex-degenerate on these models: repeated warm-started solves of the same LP flip 0.69 reactions on average between vertices of the optimal face. The LPs are solved with HiGHS (Huangfu and Hall, 2018) via `scipy.optimize.linprog`, cold-started and deterministic; a presolve-disabled retry handles the rare infeasibility mis-report. The lex optimum exists, is unique by construction, and is continuous piecewise affine in $\theta$ (Lemma 2.10); the tie-break is part of the metric's definition (Remark 2.5), and its robustness is measured in §5.6. The active-reaction census of the reference physiology is the one plain-FBA computation retained (kept for comparability with the precursor metric $\kappa_V$); it is solver-version-sensitive at the $\pm 1$-reaction level, while the gene-level panel is stable.

### 7.2 Models and data provenance

**Models.** *E. coli* iJO1366 (2,583 reactions) (Orth et al., 2011) for the association experiments of §5; iML1515 (Monk et al., 2017) for the computational batteries of §4 and the value–flux coupling verification cut. Genome-scale models are loaded from the local BiGG JSON copies with SHA-256 manifests.

**M3D microarray compendium.** The expression panels come from the M3D *E. coli* compendium v4 Build 6 (Faith et al., 2008, 2007): 4,297 gene probes × 907 arrays, $\log_2$ uniformly normalized. The carbon-depletion response is the maximum $|\log_2 \mathrm{FC}|$ over the four stationary time points (135/330/480/720 min) against the MOPS glucose reference; the oxygen-limitation and acetate-switch responses use the M9 WT anaerobic versus aerobic and MOPS acetate versus glucose contrasts. The reference-level confound control is the mean log-phase WT MOPS glucose expression. The 117 MB compendium tarball is not committed (repository size); it is re-downloadable from the recorded URL with the committed SHA-256 (`e73088f7…`).

**PRECISE.** The cross-platform arm uses the ten WT non-glucose carbon conditions of the PRECISE compendium (Sastry et al., 2019) (RNA-seq, TPM). PRECISE has no carbon-depletion class; the dissociation is reported as a negative result with per-condition values (§5.3).

**Proteomics.** Condition-dependent quantitative proteomics from Schmidt et al. (2016) (22 conditions, triplicates); integrated protein abundance from PaxDb (Wang et al., 2015); the GSE64021 (Barrett et al., 2013) time course supplies the RNA-seq carbon-starvation and the quantitative proteomics arms (§§5.8–5.9).

### 7.3 Panel construction and GPR association

The gene panel is built by reaction-based sampling: solve the lexicographic trajectory on the reference physiology (glucose 5.0 → 1.0, oxygen 22 → 5; eight anchors, 8× per-segment trajectory refinement), take the 438 reactions active ($|v| > 10^{-9}$ at any grid point) on the plain-FBA census of the reference physiology, classify each active reaction's GPR rule (single-gene 254, isozyme-OR 96, complex-AND 36, mixed 18, no-GPR 34), and assign each gene the maximum over its associated reactions. The expression panel consists of the 433 genes of this set carrying M3D response data; the primary correlation uses the 424 genes with nonzero $\kappa^\mu$. The near-colliding gene/reaction counts (424/433/435/438/440/525/537) are disambiguated in Appendix A.

### 7.4 Statistical protocols

The predictor is $\log_{10}$ of the per-gene $\kappa^\mu$ (its distribution is log-linear over five decades); the response is $\max |\log_2 \mathrm{FC}|$ of the matched contrast. Reported statistics: Pearson $r$ with analytic $p$, Spearman $\rho_S$ (raw predictor ranks, full panel and nonzero subset as labeled), partial $r$ controlling the M3D reference level by linear residualization of both variables, permutation $p$ ($10^5$ label permutations, Monte Carlo), bootstrap 95% CIs ($10^4$ resamples), and top-vs-bottom decile contrasts (one-sided Mann–Whitney $U$). All tests are two-sided except the decile contrast. The PRECISE per-condition arms use the same protocol with TPM log-fold changes. Every manuscript number traces to a deposited artifact file and is re-derived by the automated numeric verification described under Reproducibility (98 checks).

### 7.5 Reproducibility

All experiments deposit their result files and generating scripts in the public repository cited in the Data Availability statement. Independent-seed reproductions are recorded for the refinement prototype and the stress batteries. The layer-decision, path-robustness, and tie-break-robustness analyses, the cross-platform arm, and the event-measure stabilization re-run in minutes each on a single workstation. An automated numeric verification re-derives every manuscript number from the deposited artifacts (98 checks) and is re-runnable end-to-end.

## A Disambiguation of near-colliding counts

Near-colliding counts in the manuscript, each traceable to a single deposited artifact file (Data Availability):

| Value | Meaning (source) |
|---|---|
| 2,583 | iJO1366 reactions (model file; recomputed) |
| 438 | reactions active on the reference physiology, 17.0% (plain-FBA census, $\pm 1$ solver sensitivity) |
| 440 | reactions with $D_2 > 10^{-8}$ along the reference trajectory (reaction-level event count, deposited event file) |
| 433 | panel genes with M3D response (panel file row count) |
| 424 | panel genes with nonzero $\kappa^{\mu}$ — the $n$ of the primary correlation (deposited metric file) |
| 426 | nonzero genes on the P1/P2 paths (deposited path file) |
| 435 | genes with $\geq 1$ active reaction, reference physiology (census file; recomputed exactly) |
| 454 | max active reactions across the nine multi-condition panels (deposited multi-condition results) |
| 537 | union of active reactions across conditions, 20.8% (deposited multi-condition results) |
| 525 | genes with nonzero $\kappa$ across conditions (deposited multi-condition results) |
| 4 | trajectory atoms of the event measure — the four design corners carrying the full flux-layer mass 288.77; identical to the value-kink positions (deposited event files) |
| 350 | boundary edges of the $(q_{\mathrm{glc}}, q_{O_2})$ signature census, full $34 \times 34$ grid (deposited census grid) |
| 25,107 | cut events in the random-cut pool, 4,000 cuts (deposited stabilization file) |
| 1,516 | iML1515 genes, single knockouts (deposited knockout file) |
| 2,779 | double-knockout pairs, five panels (deposited knockout file) |

## B Proofs and proof sketches

### B.1 Proof of Theorem 2.11

$\Phi = c^{\top} v^*$ holds pointwise by optimality. Both sides are continuous piecewise affine (Lemma 2.10 for $v^*$; LP value function theory for $\Phi$), and differentiation of distributions is linear: applying $D^2$ to the pointwise identity gives the measure identity $D^2\Phi = \sum_r c_r \, D^2 v_r^*$. Restricting to a crease facet $F$ gives the per-crease form $\Delta\Phi'(t_k) = c^\top \Delta v'(t_k)$ at every crease time $t_k$ of any piecewise-affine trajectory, and the contraction bound $|\Delta\Phi'| \le \|c\|_\infty \|\Delta v'\|_1$ is Hölder's inequality. (i) An event with unique optima on both sides is a transversal optimal-vertex switch; its slope jump satisfies $c^\top \Delta v' = \Delta\Phi' \neq 0$ unless the value function is affine through it, which would force the optimum to persist — hence objective-invisibility ($c^\top \Delta v' = 0$, $\Delta v' \neq 0$) occurs exactly inside $\geq$ 1-dimensional optimal faces, where the tie-break selects among vertices without moving the objective; Theorem 2.11(ii) is the sparse-objective special case. The regularity input (continuity and piecewise affinity of $v^*$) is measured, not assumed: segment residuals $\leq 8 \times 10^{-14}$ (§4.1, glucose sweep) and $\Phi$ piecewise affine at $4.2 \times 10^{-13}$ (§4.4). □

### B.2 Proof of Proposition 2.12 (collapse)

If $f$ is $\lambda$-semiconvex, then $D^2 f + \lambda I \cdot \mathrm{Leb} \succeq 0$ as symmetric-matrix measures. For continuous piecewise-affine $f$, $D^2 f$ is supported on the codimension-one skeleton, hence purely singular with respect to Lebesgue; testing the inequality on skeleton sets forces the singular part itself PSD, so $D^2 f \succeq 0$. Mollifying, $D^2(f * \rho_\varepsilon) = (D^2 f) * \rho_\varepsilon \succeq 0$, and $f * \rho_\varepsilon \to f$ uniformly, so $f$ is convex as a uniform limit of convex functions. The converse is trivial. The numerical form: the smallest violating scale $h_{\max}(\lambda)$ of the OR-plus-cap value function satisfies $\lambda\, h_{\max} = 1/(2\lambda)$ exactly (AX-10), i.e. no finite semiconvexity constant exists at any scale. The signed-layer statement is elementary distribution theory: a continuous PWL function has distributional $D^2$ supported on its crease skeleton, a finite symmetric Radon measure with per-crease sign flags. □

### B.3 Proof of Proposition 2.13

At a generic codimension-two vertex of a convex PWL $f$ on $\mathbb{R}^2$, exactly three facets meet with gradients $n_1, n_2, n_3$ (cyclically ordered). The Monge–Ampère atom is the volume of the normal fan $\mathrm{conv}\{n_i\}$ (the gradient image sweeps the triangle). Writing $j_i = n_{i+1} - n_i$ for the codimension-one jumps, twice the fan area is $|\det(j_1, j_2)|$ (the shoelace formula on the triangle), giving atom $= \frac{1}{2}|\det(j_1, j_2)| = \frac{1}{2}|j_1||j_2| \sin\angle$; the product $|j_1||j_2|$ overestimates by $2/\sin\angle \geq 2$ with equality iff $j_1 \perp j_2$. The simplex-fan generalization to dimension $p$ is the same shoelace argument on a $p$-simplex fan. Machine verification: 400 random max-of-affine vertices, determinant = fan area to $8.9 \times 10^{-16}$, the $2/\sin\angle$ law to $2.5 \times 10^{-12}$, orthogonal edge case exactly 2. □

## Data, Software, and Code Availability

All figures, tables, and numerical claims in this manuscript are produced by committed scripts from deposited artifacts; the complete bundle (generation scripts, result artifacts, genome-scale models, data manifests with SHA-256 checksums, and the verification scripts that re-derive every manuscript number) is available at github.com/ MIKEAA2020/deepseek-highly-general. Genome-scale models (iJO1366, iML1515) were obtained from the BiGG database; expression compendia from M3D and PRECISE; proteome data from PaxDb and the condition-dependent proteome of Schmidt et al.; all sources are public and cited in the reference list. No new experiments were performed.

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## Competing Interests

The author declares no competing interests, financial or otherwise, in relation to the work described in this manuscript.

## Ethics Approval

Not applicable. The study is entirely computational: it involves no human participants, human tissue, or animal subjects; it performs no new experiments; and it analyzes only mathematical models of metabolic networks together with previously published, publicly available datasets.

## Declaration of Generative AI in Scientific Writing

During the preparation of this work the author used GLM (Z.ai) and DeepSeek large language models, under the author's direction and supervision, to assist with manuscript drafting and editing and with the development and documentation of the analysis code. All AI-assisted content was reviewed, verified, and edited by the author, who takes full responsibility for the final content of the manuscript and for the integrity of its results.

## Author Contributions

X is the sole author of this manuscript. X conceived the study design, developed the methodology and analysis code, performed the data analysis, and drafted and revised the manuscript, and approved the submitted version.

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